UTrucking AI Phone Assistant — Build Plan, Progress & Roadmap

Last updated: 2026-07-02 · **Live agent:** v33 · **Backend:** commit `282d7b0` (deployed)

Where we are today

The assistant is **live and fully tested**. It answers calls in a warm, natural voice, looks up any student's order across **both** data sheets, verifies the caller's identity before sharing anything, answers questions one at a time, handles general questions from a knowledge base, and transfers to the team on request. As of 2026-07-02 the backend is fixed and redeployed: the item/invoice sheet now loads (654 records) and the name-match cutoff was tightened to reduce false matches.

Delivered so far

Capability	Status
Order lookup by name (handles mispronounced / misspelled names)	✅ Live
Natural, one-answer-at-a-time conversation (no rambling)	✅ Live
Identity verification — name **plus** a second detail	✅ Live
Privacy protection — never reveals another student's data	✅ Live
Two-sheet join (dispatch logistics + invoice/item detail)	✅ Live
Pricing / service questions from the knowledge base	✅ Live
Transfer to the office at (314) 266-8878	✅ Live
Safety guardrails (stays on-task, resists misuse)	✅ Live
Post-call logging (found, order ID, resolution, sentiment)	✅ Live
Hard-name + impostor stress testing	✅ Done

Capability	Status
Documentation: flow maps, connections, QA log, exec deck, data audit	✅ Done

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Roadmap — from *answering* to *doing*

Each future item below lists **what it entails**, a **build-time estimate**, **what it requires (with real costs)**, and how it **future-proofs** the operation. Estimates assume one developer and the current Render + Google Sheets stack.

Phase 1 — Information assistant · ✅ DONE (live)

Answers questions, verifies identity, transfers to a human. Deflects the highest-volume, repetitive calls.

Phase 2 — The receptionist replacement · 🕓 NEXT

2a. Schedule / reschedule / cancel a pickup on the call

- **What it entails:** give the agent write-access tools so it can create, move, or cancel a booking during the call, then read back a confirmation. Requires a "source of truth" for availability (time-slot capacity per day).
- **Build time:** ~1–2 weeks (backend write endpoints + 3 Retell tools + confirmation logic + guardrails against double-booking).
- **Requires:** a **Google service account** (free) so the backend can *write* to the sheet — public CSV is read-only; or a small database. No new per-call cost.
- **Future-proofs:** turns the bot from "answers about orders" into "books orders" — the single biggest staff-time saver.

2b. Auto-identify the caller by phone number

- **What it entails:** read Retell's inbound `from_number`, match it to the **Phone** column already in the DISPATCH sheet, and greet the known caller by name — most people never have to spell anything.
- **Build time:** ~2–3 days.

- **Requires:** a **provisioned phone number** (Retell/Twilio ≈ **$1–3 / month** for the number) + voice usage (**≈ $0.07–0.10 / minute** all-in for STT + LLM + TTS). Budget a one-time **~$20** for number setup + first-month testing.
- **Future-proofs:** faster, friendlier calls; foundation for outbound reminders (Phase 3).

2c. Callback / message capture (after-hours)

- **What it entails:** when the office is closed or a transfer fails, take a message (name, number, reason) and log it for the team.
- **Build time:** ~2 days.
- **Requires:** a place to drop messages (a sheet tab or an email via a free SendGrid tier). ~$0.
- **Future-proofs:** zero missed leads outside business hours.

Phase 3 — Scale & growth · 🔭 LATER

3a. New-customer registration (sign up on the call)

- **Entails:** capture a new storage sign-up (name, dorm, dates, service) and write it to the intake sheet. **Build:** ~1 week. **Requires:** write-back (2a) + validation. **Future-proofs:** the bot drives *revenue*, not just service.

3b. Text (SMS) confirmations & booking links

- **Entails:** text order details or a payment/booking link after the call. **Build:** ~3 days. **Requires: Twilio** (~$1/mo number + **~$0.0079 per SMS**) or Retell SMS. **Future-proofs:** fewer no-shows, written proof for the customer.

3c. Team notifications (email / Slack)

- **Entails:** post-call summaries pushed to staff. **Build:** ~2 days. **Requires:** Slack webhook (free) or SendGrid free tier. **Future-proofs:** the team sees every call without listening to recordings.

3d. Outbound reminders ("your pickup is tomorrow")

- **Entails:** scheduled outbound calls/texts the day before. **Build:** ~1 week. **Requires:** a scheduler + outbound voice/SMS budget. **Future-proofs:** cuts missed pickups (a direct cost — see the data audit).

3e. Spanish for parents · analytics dashboard

- **Entails:** enable multilingual voice; stand up a live metrics dashboard from call + order data. **Build:** ~2 days (language) / ongoing (dashboard — this report is step one). **Requires:** ~$0 extra. **Future-proofs:** reach + data-driven decisions.

From audit to action — what the AI assistant itself will do

The **Data & Revenue Audit** lists six moves. Here is exactly **who runs each**, and — for the ones the assistant runs — whether it can do it **today** or **next (Phase 2, once booking is enabled).**

Audit recommendation	On the AI assistant?	Status
Quote consistent, correct pricing (incl. a new box price)	Yes	✅ **Already does this** — answers from the knowledge base
Confirm the caller's building / room (clean-data first step)	Yes	✅ **Already does this** — part of the identity gate
Smooth the peak — steer callers to open shoulder days	Yes	🕓 Next — needs scheduling (Phase 2a)
Upsell containers & fridges at booking	Yes	🕓 Next (Phase 2a)
Close billing leakage — no booking without a non-zero invoice	Yes + backend rule	🕓 Next (Phase 2a)
Fix data at the source — capture clean building + phone at booking	Yes	🕓 Next (Phase 2a)
Raise the box price $2	No — management decision	👤 Business sets it; the agent then quotes it
Clear the not-dispatched backlog	No — dispatch / ops team	👥 Operations, not a phone task

The four the AI caller itself will do

All four are unlocked by the **same Phase 2 booking capability** — build it once and the assistant executes all of them:

1. **Smooth the peak.** When a caller asks for a slammed day (e.g., May 12), the assistant offers the nearest **open** day/time first — "I have 9 AM on the 15th, or a short wait on the 12th" — spreading load off the two peak days that hold **74%** of the season's revenue. *Needs: booking + a per-day capacity check.*
2. **Upsell the basket.** After the core order, it offers the add-ons the data shows people actually take — **Plastic Container ($18)**, **Mini Fridge ($23)** — nudging basket size above the ~7-item average. *Needs: booking + item catalog.*
3. **Close the leak.** The assistant (plus a backend rule) will not finalize a storage booking with a **$0 total or missing invoice** — the exact gap that cost **~$1,056** this season. *Needs: booking + a validation rule.*
4. **Clean the data at the source.** At booking it confirms and records the **correct building** and a **working phone number** — killing the 80 "unknown building" rows and enabling caller-ID recognition next season. *Needs: booking write-back.*

What it already does today (no new work)

- **Quotes pricing consistently** from the knowledge base — so a box-price increase needs **zero** agent changes.
- **Confirms building / room** on every verified call as part of the identity gate — the first step of clean-data-at-the-source.

The remaining two moves are **not** phone tasks: setting the price is a management decision, and clearing the dispatch backlog belongs to the ops team.

Expansion roadmap — new AI tools & automations

Beyond the phone assistant, the **same foundation** (Retell + the Render backend + your sheets) can power a full suite of tools. We build in three waves — **A → B → C**, one tool at a time, each tested against real data before the next. **Wave D is logged for later.** *(Impact/effort are directional; dollar figures come from the Data & Revenue Audit.)*

Wave A · Win more bookings — revenue 🟢 building first

Tool	What it does	Why it matters
Photo-to-quote	Student photographs their pile; AI counts items and returns an instant itemized quote	You already collect item photos + invoices — a unique data moat; drives conversion & upsell
Web + SMS booking assistant	The phone agent's brain, on text and web chat	Students text, they don't call; reuses the existing backend
Spanish parent line	Multilingual voice	Parents handle logistics & payment
Group / referral booking	Roommates / floors book together	Dorms already cluster — cuts cost per stop

Wave B · Survive the peak — operations 🟢 after A

Tool	What it does	Why it matters
Smart scheduler	Steers callers to open shoulder-day slots	Protects the 74%-in-5-days revenue from capacity loss
Reminders & confirmations	"Your pickup is tomorrow — reply to reschedule"	Cuts no-shows, pre-fills the schedule
Auto-dispatch + routing	Clusters orders by building, sequences stops, assigns crews	Attacks the 212-order not-dispatched backlog
Movers' field app	Crew route + item lists + photo/complete, auto-writes back to the sheet	Ends manual data entry & "unknown building"

Wave C · Stop leaking money — finance 🟢 after B

Tool	What it does	Why it matters
Invoice automation + leakage guard	Auto-invoice; block $0 / missing-invoice orders	Recovers the ~$1,056 gap; clean books
Payment chaser	SMS follow-up with a pay link	Faster cash
Damage / condition vision docs	Auto-tag item condition from the photos already taken	Dispute protection + a protection-plan upsell

Wave D · See & predict — intelligence 🔭 logged for later

Live ops dashboard · demand forecast (per-building, keyed to the academic calendar) · "ask-your-data" staff copilot · fall **return-season** automation (Rental Returns = 25% of volume).

Build order & method. A → B → C, one tool at a time; each is built, tested against real data, then wired into the live agent. Detailed engineering steps and the exact third-party accounts required are tracked in a separate technical plan (kept out of this report).

Future-proofing audit (do these before scaling)

Small investments now that prevent expensive breakage later:

Area	Today	Risk if left	Recommended before scale
Deploys	Render auto-deploy is **OFF**; deploys are manual	A fix silently never ships	Re-enable auto-deploy **or** wire a deploy hook; document the one-button process
Data store	Google Sheets read as CSV per call	Sheets are fine for reads to a few-thousand rows, but **can't handle concurrent writes** (needed for booking)	Move to **Postgres on Render** (free/$7 tier) when Phase 2a lands
Uptime	Render free tier spins down → cold-start delay on first call	First caller of the hour waits ~30–60s	`keep_alive` helps; a **paid Render instance (~$7/mo)** removes cold starts
Backend access	Endpoints are open (`CORS *` , no auth)	Anyone can call `lookup_student`	Add a shared secret/header check
Join key	Name-based join across two sheets	Multi-order customers can mix records	Join on **`order_id`**
Backups	Live sheets only	Accidental edit = data loss	Nightly export/backup of both sheets
Secrets	API keys in local config	—	Keep in Render env vars; rotate periodically

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What each phase unlocks

- **Phase 1** deflects the highest-volume, repetitive calls — the front desk stops answering the same questions.
- **Phase 2** lets the assistant *do* the receptionist's work (booking, rescheduling, caller ID) — the biggest staff-time win.
- **Phase 3** drives growth (sign-ups), proactive service, and multilingual reach.

Dependencies & open items

- Phase 2 / 3 backend features deploy to the Render service (**manual deploy** until auto-deploy is re-enabled).
- **Provision a phone number** in Retell when ready to go live on a real line (~$20 to start).
- Optional: **restore any missing detail** in the SERVICE sheet (item lists are sparse on some rows — see the data audit).